

Day 2: Global Infrastructure, Shared Responsibility and Account Safety

Sunday, 11 October 2026 | 7:00-9:00 PM IST | Cloud Unlock x AWS Student Builder Group, MIT ADT University | Trainer: Gangadhar Ure

1. Goal of Today

Understand where AWS physically lives (Regions, Availability Zones, edge locations), who is responsible for what (shared responsibility model), and lock down your account so a mistake does not cost you money or data.

You will learn	You will do
Region, Availability Zone, edge location	Explore Regions and AZs in Console and CLI
How to choose a Region	Pick a Region for a sample app using 4 criteria
Shared responsibility model	Sort 12 tasks: AWS or customer?
Root user dangers	Turn on MFA for the root user
Why billing alerts matter	Create a budget alert

2. Before Class Checklist

- Day 1 account is working and you can sign in.
- An authenticator app on your phone (Google Authenticator, Microsoft Authenticator, Authy or similar).
- Your root user email and password (you will need them once, today).
- Homework done: you can list 10 services.

3. AWS Global Infrastructure

Term	Meaning	Easy example
Region	A geographic area with multiple isolated locations	Mumbai (ap-south-1), Hyderabad (ap-south-2)
Availability Zone (AZ)	One or more data centres in a Region, with separate power and networking, linked by fast low-latency links	ap-south-1a, ap-south-1b
Edge location / PoP	Small sites close to users used to cache content (CloudFront)	Delivering your video fast to a user in Pune
Local Zone	Extension of a Region closer to a big city for very low latency	Gaming or live video near a city
Wavelength Zone	AWS compute inside 5G carrier networks	Ultra-low latency mobile apps

Note: AWS publishes live counts on aws.amazon.com/about-aws/global-infrastructure. At the time of writing the page lists about 39 Regions, 124 Availability Zones and 750+ CloudFront points of presence. Numbers grow, so check the live page and never memorise counts for the exam. The Hyderabad Region is opt-in and must be enabled before use.

Best practice: Design for high availability by spreading across at least two Availability Zones. One AZ failing should not take your app down.

How to choose a Region (exam favourite)

Factor	Question to ask
Compliance / data residency	Does the law or customer require data to stay in India?
Latency	Where are most of my users? Closer is faster.
Service availability	Is the service/feature I need available in that Region?
Cost	Prices differ by Region

Exam shortcut: Data must stay in a country = choose a Region in that country. Users far away = use edge locations/CloudFront. High availability = multiple AZs. Disaster recovery from a Region-wide issue = multiple Regions.

4. Practical 1: Region and AZ Explorer (about 20 min)

- 1 Open the Console and click the Region selector. Count how many Regions are enabled for your account.
- 2 Open Account settings and scroll to AWS Regions. Find Asia Pacific (Hyderabad). Note whether it is enabled or disabled. Do NOT enable it today unless asked.
- 3 Open CloudShell in Mumbai and run the commands below.
- 4 Open the EC2 service > 'Launch instance' page in Mumbai and look at the Availability Zone choices in the Network settings subnet list. DO NOT launch. Close the page.

```
# Regions enabled for your account
aws ec2 describe-regions --query "Regions[].RegionName" --output text

# All Regions including disabled ones
aws ec2 describe-regions --all-regions --query "Regions[][RegionName,OptInStatus]" --output table

# AZs in Mumbai
aws ec2 describe-availability-zones --region ap-south-1 --query "AvailabilityZones[][ZoneName,ZoneId,State]" --output table

# AZs in N. Virginia, for comparison
aws ec2 describe-availability-zones --region us-east-1 --query "AvailabilityZones[][ZoneName]" --output text
```

Expected result: Mumbai shows 3 or more AZs (read your own output). Opt-in Regions show 'not-opted-in'. Different Regions have different numbers of AZs.

Note: The zone NAME (ap-south-1a) is mapped differently per account. The zone ID (aps1-az1) is the same for everyone. That is why AWS shows both.

5. Shared Responsibility Model

	AWS is responsible for	Customer is responsible for
Headline	Security OF the cloud	Security IN the cloud
Examples	Data centres, hardware, host OS, virtualisation, global network	Your data, IAM users and permissions, firewall rules, OS patching on EC2, encryption choices

Responsibility shifts by service

Service	AWS handles	You handle
Amazon EC2 (IaaS)	Hardware and virtualisation	Guest OS, patches, apps, security groups, data
Amazon RDS (managed)	Hardware, OS and database patching	Data, access control, network access, parameter settings
AWS Lambda (serverless)	Servers, OS, runtime	Your code, permissions, data
Amazon S3	Hardware and service durability	Bucket permissions, encryption choice, what you store

Activity (10 min): Sort these 12 tasks into AWS or Customer and write the letter A or C. Answers are at the end of the page.

#	Task	A / C
1	Replacing a failed disk in a data centre	
2	Deciding who can read your S3 bucket	
3	Physical security of the building	
4	Patching the OS on your EC2 instance	
5	Patching the database engine on RDS	
6	Encrypting your customer data	
7	Keeping the global network cables running	
8	Rotating your IAM user passwords	
9	Configuring your security group rules	
10	Keeping the Lambda runtime up to date	
11	Deciding which Region stores your data	
12	Hypervisor security	

Answers: 1-A, 2-C, 3-A, 4-C, 5-A, 6-C, 7-A, 8-C, 9-C, 10-A, 11-C, 12-A

Exam shortcut: If the question mentions physical, hardware, hypervisor or global infrastructure, the answer is AWS. If it mentions your data, your users, your permissions or your OS on EC2, the answer is customer.

6. Practical 2: Secure Your Root User (about 20 min)

The root user is the email/password you used to create the account. It has unrestricted access that cannot be limited. Treat it like the master key to a building.

Guardrail: Never create access keys for the root user. Never use root for daily work. Do not record or share your QR code, secret key or backup codes.

- 1 Sign in as the root user. Open the account menu (top right) > Security credentials.
- 2 Under Multi-factor authentication (MFA), choose Assign MFA device. Name it 'root-mfa-phone'.
- 3 Choose Authenticator app. Scan the QR code with your authenticator app (do NOT screenshot the QR).
- 4 Enter two consecutive codes from the app. Finish.
- 5 Sign out. Sign in again as root and confirm it asks for the MFA code.

- 6 On the same Security credentials page, confirm there are NO root access keys. If you see any, delete them.

Expected result: The root user shows an assigned MFA device and zero access keys.

7. Practical 3: Budget Alert (about 15 min)

Note: AWS Budgets sends alerts. It does NOT stop spending. A budget is a smoke alarm, not a sprinkler.

- 1 Open Billing and Cost Management > Budgets > Create budget.
- 2 Choose a template such as 'Zero spend budget' or 'Monthly cost budget'. If neither is shown, choose a custom Cost budget.
- 3 Name it 'cloud-unlock-learning-budget'. Set the amount to USD 5 if asked for an amount.
- 4 Enter your personal email for alerts. Create the budget.
- 5 Open Billing > Free Tier (or Credits) and note your remaining credits again.

Best practice: Set a budget on day one, tag everything you create, and delete lab resources the same day.

8. Evidence Checklist

- Screenshot: root MFA shown as assigned (no QR, no codes, no account ID).
- Screenshot: the budget list showing 'cloud-unlock-learning-budget'.
- Your Mumbai AZ output with the account ID not visible.
- Your answers to the 12-task card sort.

9. Quick Quiz (answers below)

- 1 What is an Availability Zone? A) A country B) One or more data centres in a Region C) A CDN edge D) A billing group
- 2 Which service lowers latency by caching content close to users? A) CloudFront B) IAM C) RDS D) KMS
- 3 Under shared responsibility, who patches the guest OS on EC2? A) AWS B) Customer C) Both D) Neither
- 4 Best practice for the root user? A) Create access keys B) Enable MFA and avoid daily use C) Share the password D) Use it for all tasks
- 5 Does AWS Budgets stop spending automatically at the limit? A) Yes B) No, it sends alerts C) Only in Mumbai D) Only on Free plan

Answers: 1-B, 2-A, 3-B, 4-B, 5-B

10. Homework (30 min)

- Write a 4-line Region choice for a fest-registration website for students in Pune (use the four factors).
- Memorise the shared responsibility sentence: security OF the cloud vs IN the cloud.
- Day 3 is IAM: users, groups, roles and policies. Read about 'least privilege' on the AWS IAM page.

11. Cleanup

Keep MFA and the budget (they are free). If you opened any pages for demo purposes, close them. No resources to delete.

12. References

- AWS Certified Cloud Practitioner (CLF-C02) Exam Guide, Domain 2: Security and Compliance and Domain 1: Cloud Concepts.
- AWS Global Infrastructure: aws.amazon.com/about-aws/global-infrastructure
- AWS Shared Responsibility Model: aws.amazon.com/compliance/shared-responsibility-model
- AWS Budgets documentation: docs.aws.amazon.com/cost-management